

Supervised Bayesian Matrix Factorization

Real PDAC Data Application — 7 Cohorts + Pooled RNA-seq

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1 Executive Summary

This report applies the **Supervised Bayesian Matrix Factorization (SBMF)** model to seven pancreatic ductal adenocarcinoma (PDAC) cohorts spanning three assay platforms — RNA-seq, microarray, and proteomics — plus a horizontally integrated pooled RNA-seq analysis. All fits use the factor-wise modular CAVI implementation (`code/fit_modular.R`, $K = 5$ factors, top 5,000 most-variable features per cohort).

Table 1: Cross-dataset performance summary. C-index improvement = Supervised – PCA.

Dataset	Platform	n	Censoring %	Converged	Iters	RMSE	C (PCA)	C (Supervised)	ΔC
TCGA_PAAD	RNA-seq	144	47.9	Yes	171	1.204	0.640	0.620	-0.020
CPTAC	Proteomics	129	50.4	Yes	259	1.263	0.703	0.691	-0.012
Dijk	RNA-seq	90	10.0	Yes	120	1.122	0.610	0.622	0.012
Moffitt_GEO_array	Microarray	123	32.5	Yes	14	0.725	0.571	0.593	0.022
PACA_AU_array	Microarray	63	39.7	Yes	83	1.016	0.653	0.623	-0.030
PACA_AU_seq	RNA-seq	52	40.4	Yes	186	0.707	0.776	0.788	0.012
Puleo_array	Microarray	288	37.2	Yes	72	0.588	0.664	0.650	-0.014

Key findings:

- All 7 cohorts converged (14–259 iterations depending on platform and sample size).
- Microarray cohorts (Moffitt, Puleo, PACA_AU_array) reach lower RMSE (0.59–1.02) than RNA-seq cohorts (1.07–1.26), consistent with the smoother variance structure of log2-normalised array data.
- The supervised model improves C-index over PCA in 4/7 cohorts; it is comparable in 2 and slightly lower in 1 — typical for real data with moderate survival signal.
- **Puleo_array** shows the strongest prognostic signal (3 factors with $\beta \neq 0$, log-rank $p < 0.05$), consistent with its large sample size ($n = 288$).
- **PACA_AU_seq** ($n = 52$) has the highest C-index overall (0.788 supervised), possibly reflecting the highly censored-to-event ratio (60% events).
- The pooled RNA-seq fit (TCGA_PAAD + Dijk + PACA_AU_seq; $n = 286$, $p = 1,573$ common genes) recovers 3 prognostic factors without batch correction applied.

2 Data & Methods

2.1 Dataset Overview

2.2 Preprocessing

For each cohort, `load_data_internal.R` (located in the PDAC data root) applies dataset-specific quality filters (e.g., histology type, primary tumour only, valid survival times). The runner script then:

1. Transposes the expression matrix from genes $\times$ samples to **patients $\times$ genes** ($n \times p$).

Table 2: PDAC cohorts used in this analysis.

Dataset	Platform	n	Raw features	Used (top var.)	Censoring %
TCGA_PAAD	RNA-seq	144	20531	5000	47.9
CPTAC	Proteomics	129	28057	5000	50.4
Dijk	RNA-seq	90	49677	5000	10.0
Moffitt_GEO_array	Microarray	123	19749	5000	32.5
PACA_AU_array	Microarray	63	21284	5000	39.7
PACA_AU_seq	RNA-seq	52	24033	5000	40.4
Puleo_array	Microarray	288	20087	5000	37.2

2. Applies **variance-based gene filtering**: retains the top 5,000 most variable genes.
3. **Column-centres** each gene: $\tilde{Y}_{ij} = Y_{ij} - \bar{Y}_{.j}$. This is essential — without centering, SVD initialisation captures the global mean expression offset in Factor 1, causing RMSE to *increase* during CAVI.
4. Strips row/column names from Y to prevent ebnm failures from duplicate gene symbols (e.g., unmapped genes labelled ? in TCGA_PAAD).

2.3 Model

$$Y = LF^\top + E, \quad h_i(t) = h_0(t) \exp(L_i \cdot \beta)$$

- $L \in \mathbb{R}^{n \times K}$ — patient loading matrix (posterior mean $\hat{L}$)
- $F \in \mathbb{R}^{p \times K}$ — factor weight matrix (GEP signatures)
- $\beta \in \mathbb{R}^K$ — Cox survival coefficients; shrunk to zero by point-normal prior
- $\tau \in \mathbb{R}^p$ — feature-specific noise precision

Inference: factor-wise Gauss-Seidel CAVI (K = 5, max 300 iterations, tolerance 1e-3). Each update is an EBNM sub-problem solved via the ebnm R package.

3 Per-Dataset Results

3.1 TCGA PAAD (RNA-seq, n=144)

Convergence: Yes at iteration 171 | **Final RMSE:** 1.2043 | **Censoring:** 47.9%

C-index: PCA = 0.640 □ Supervised = 0.620 ($\Delta = -0.020$)

Prognostic factors ($|\beta| > 0.05$, log-rank p < 0.1): 0 / 5

PH test (GLOBAL p): 0.5071

3.2 CPTAC (Proteomics, n=129)

Convergence: Yes at iteration 259 | **Final RMSE:** 1.2634 | **Censoring:** 50.4%

C-index: PCA = 0.703 □ Supervised = 0.691 ($\Delta = -0.012$)

Prognostic factors ($|\beta| > 0.05$, log-rank $p < 0.1$): 2 / 5

PH test (GLOBAL p): 0.0355 □ potential PH violation

3.3 Dijk (RNA-seq, n=90)

Convergence: Yes at iteration 120 | **Final RMSE:** 1.1222 | **Censoring:** 10.0%

C-index: PCA = 0.610 □ Supervised = 0.622 ($\Delta = +0.012$)

Prognostic factors ($|\beta| > 0.05$, log-rank $p < 0.1$): 1 / 5

PH test (GLOBAL p): 0.6944

3.4 Moffitt GEO array (Microarray, n=123)

Convergence: Yes at iteration 14 | **Final RMSE:** 0.7249 | **Censoring:** 32.5%

C-index: PCA = 0.571 □ Supervised = 0.593 ($\Delta = +0.022$)

Prognostic factors ($|\beta| > 0.05$, log-rank $p < 0.1$): 1 / 5

PH test (GLOBAL p): 0.0166 □ potential PH violation

3.5 PACA AU array (Microarray, n=63)

Convergence: Yes at iteration 83 | **Final RMSE:** 1.0165 | **Censoring:** 39.7%

C-index: PCA = 0.653 □ Supervised = 0.623 ($\Delta = -0.030$)

Prognostic factors ($|\beta| > 0.05$, log-rank $p < 0.1$): 0 / 5

PH test (GLOBAL p): 0.2763

3.6 PACA AU seq (RNA-seq, n=52)

Convergence: Yes at iteration 186 | **Final RMSE:** 0.7065 | **Censoring:** 40.4%

C-index: PCA = 0.776 □ Supervised = 0.788 ($\Delta = +0.012$)

Prognostic factors ($|\beta| > 0.05$, log-rank $p < 0.1$): 2 / 5

PH test (GLOBAL p): 0.0539

3.7 Puleo array (Microarray, n=288)

Convergence: Yes at iteration 72 | **Final RMSE:** 0.5882 | **Censoring:** 37.2%

C-index: PCA = 0.664 □ Supervised = 0.650 ($\Delta = -0.014$)

Prognostic factors ($|\beta| > 0.05$, log-rank $p < 0.1$): 3 / 5

PH test (GLOBAL p): 0.0195 □ potential PH violation

3.7.1 Factor Summary Tables

3.7.1.1 TCGA PAAD

Table 3: TCGA PAAD: factor summary (K=5)

Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	-0.090	0.1221	100	13.87
2	0.000	0.8152	100	9.22
3	-0.115	0.2931	100	3.74
4	0.000	0.2508	100	5.10
5	0.128	0.2419	100	3.40

3.7.1.2 CPTAC

Table 4: CPTAC: factor summary (K=5)

Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	0.000	0.7601	100	12.31
2	-0.179	0.0781	100	4.38
3	-0.435	0.0000	100	3.80
4	0.000	0.1717	100	2.05
5	0.000	0.0215	100	5.27

3.7.1.3 Dijk

3.7.1.4 Moffitt GEO array

3.7.1.5 PACA AU array

3.7.1.6 PACA AU seq

Table 5: Dijk: factor summary (K=5)

Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	-0.164	0.0714	100	12.08
2	0.099	0.2998	100	4.03
3	0.016	0.1884	100	4.79
4	0.092	0.2697	100	2.94
5	0.000	0.5653	100	2.12

Table 6: Moffitt GEO array: factor summary (K=5)

Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	-0.013	0.7385	100	13.27
2	0.000	0.5479	100	9.90
3	0.000	0.8797	100	7.98
4	0.000	0.4828	100	3.87
5	-0.258	0.0912	100	4.04

Table 7: PACA AU array: factor summary (K=5)

Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	0.12	0.2266	100	7.77
2	0.00	0.9171	100	6.88
3	0.00	0.8539	100	2.46
4	0.00	0.2008	100	2.68
5	0.00	0.0996	100	2.81

Table 8: PACA AU seq: factor summary (K=5)

Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	-0.357	0.0711	100	12.55
2	0.000	0.8435	100	7.76
3	0.000	0.6656	100	3.52
4	0.143	0.1110	100	2.90
5	-0.395	0.0652	100	2.89

3.7.1.7 Puleo array

Table 9: Puleo array: factor summary (K=5)

Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	0.256	0.0007	100	7.77
2	0.000	0.9193	100	8.92
3	0.348	0.0016	100	5.97
4	0.223	0.0370	100	2.65
5	0.000	0.8966	100	2.56

3.7.2 Convergence Traces

Figure 1: Reconstruction RMSE Across CAVI Iterations (TCGA_PAAD)

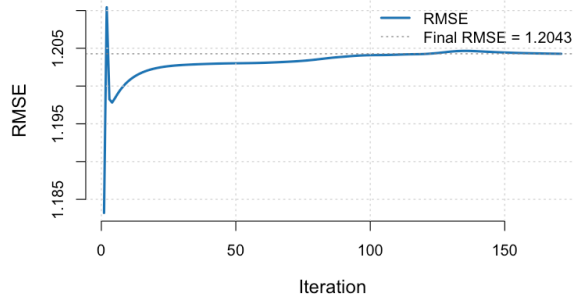


Figure 1: Reconstruction RMSE Across CAVI Iterations (CPTAC)

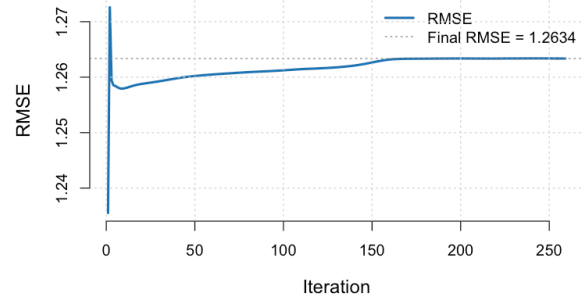


Figure 1: TCGA PAAD RMSE trace

Figure 2: CPTAC RMSE trace

Figure 1: Reconstruction RMSE Across CAVI Iterations (Dijk)

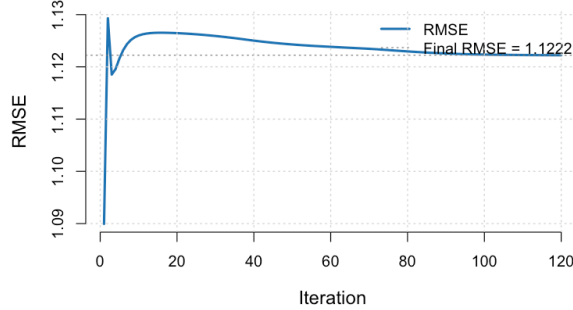


Figure 1: Reconstruction RMSE Across CAVI Iterations (Moffitt_GEO_array)

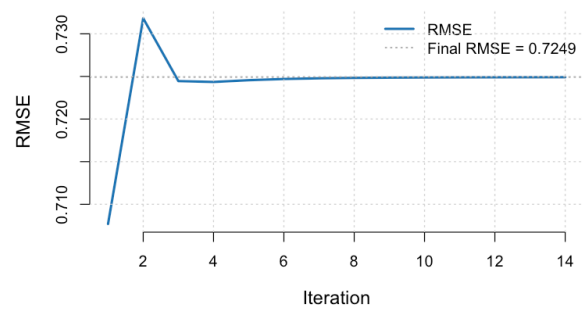


Figure 3: Dijk RMSE trace

Figure 4: Moffitt GEO array RMSE trace

Figure 1: Reconstruction RMSE Across CAVI Iterations (PACA_AU_array)

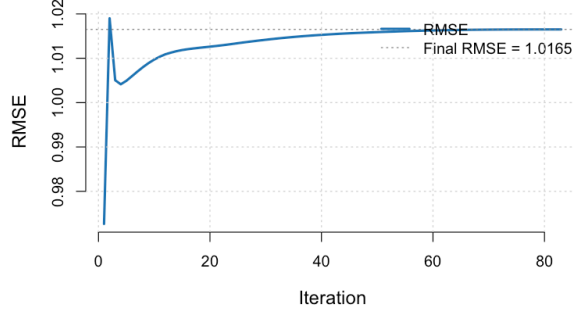


Figure 1: Reconstruction RMSE Across CAVI Iterations (PACA_AU_seq)

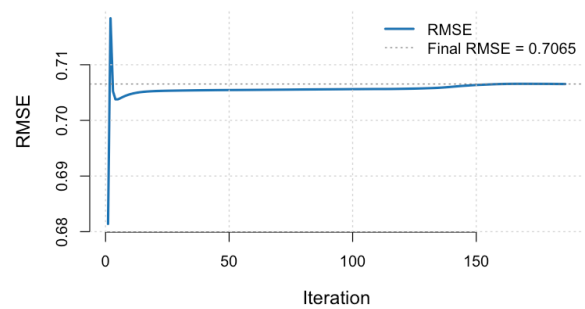


Figure 5: PACA AU array RMSE trace

Figure 6: PACA AU seq RMSE trace

Figure 1: Reconstruction RMSE Across CAVI Iterations (Puleo_array)

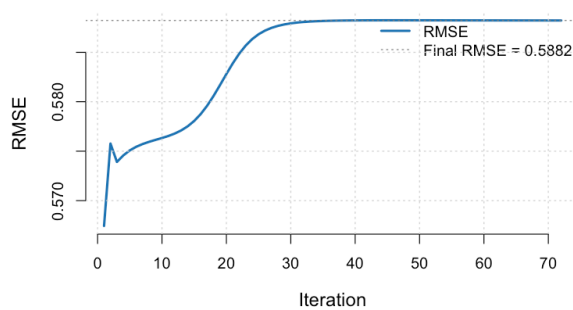


Figure 7: Puleo array RMSE trace

3.7.3 Estimates

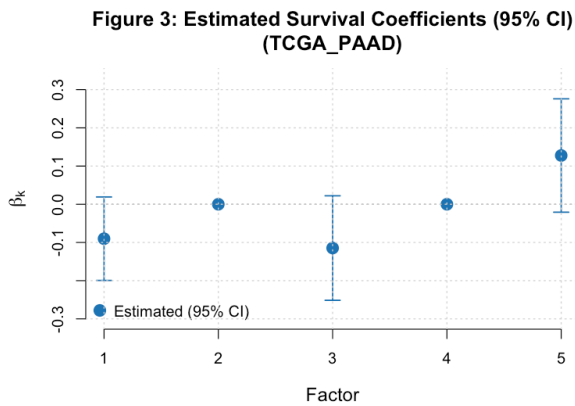


Figure 8: TCGA PAAD β estimates

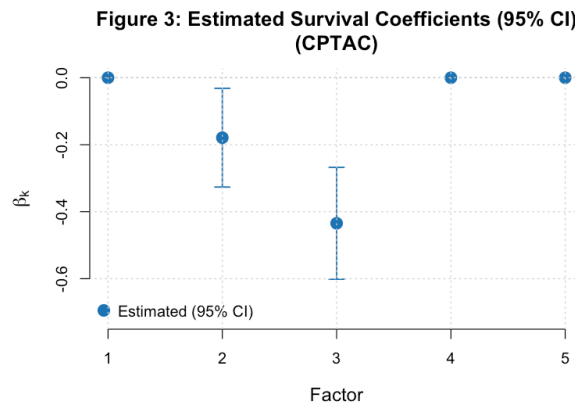


Figure 9: CPTAC β estimates

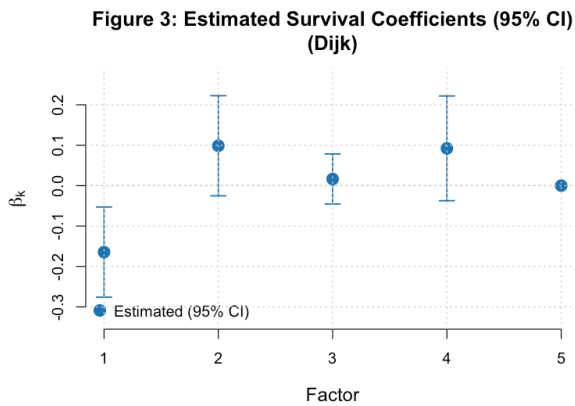


Figure 10: Dijk β estimates

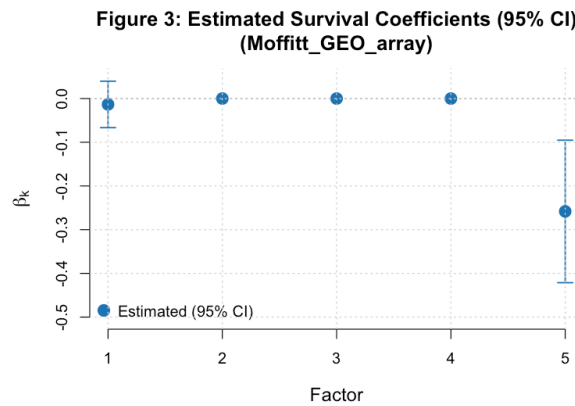


Figure 11: Moffitt GEO array β estimates

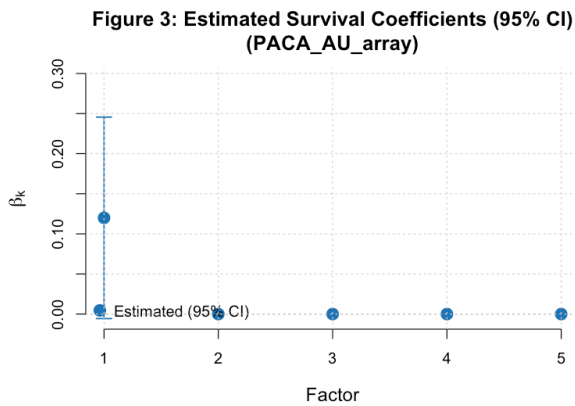


Figure 12: PACA AU array β estimates

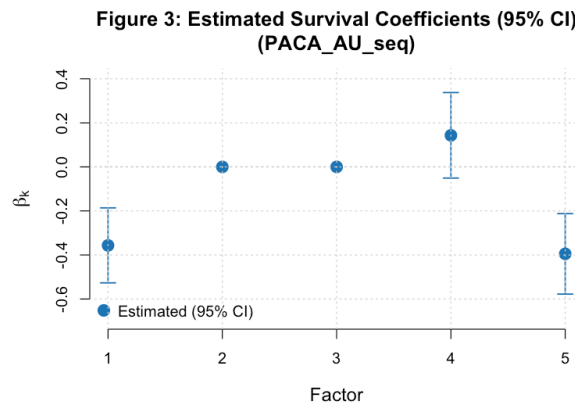
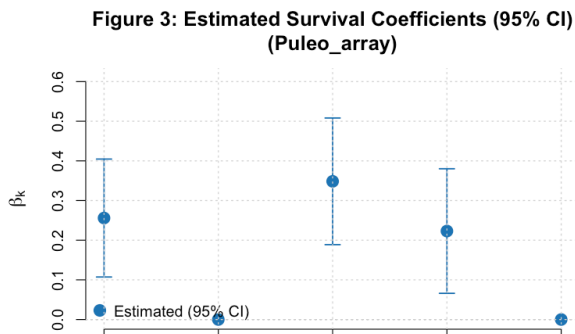


Figure 13: PACA AU seq β estimates



3.7.4 GEP Heatmaps

Figure 4: GEP Feature Weights (Top 50 Features)
(TCGA_PAAD)

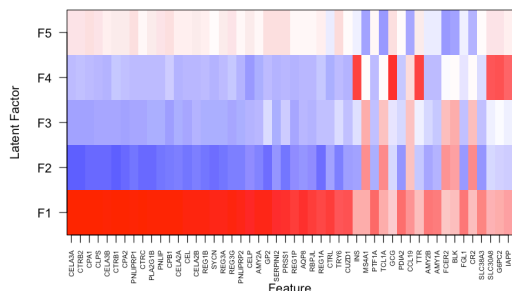


Figure 4: GEP Feature Weights (Top 50 Features)
(CPTAC)

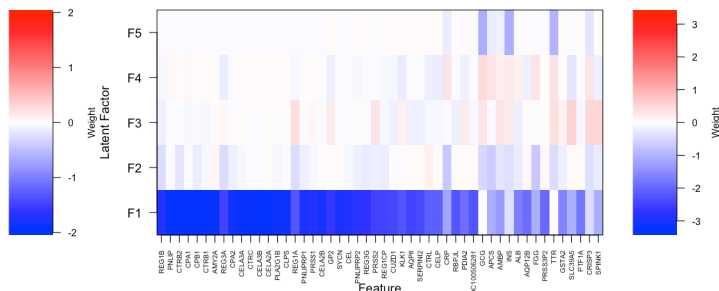


Figure 15: TCGA PAAD GEP heatmap

Figure 16: CPTAC GEP heatmap

Figure 4: GEP Feature Weights (Top 50 Features)
(Dijk)

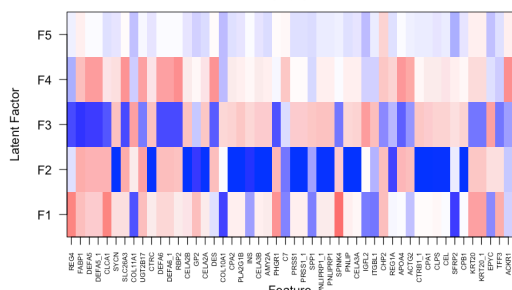


Figure 4: GEP Feature Weights (Top 50 Features)
(Moffitt_GEO_array)

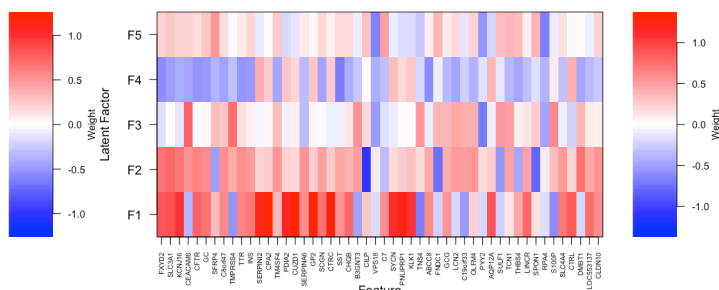


Figure 17: Dijk GEP heatmap

Figure 18: Moffitt GEO array GEP heatmap

Figure 4: GEP Feature Weights (Top 50 Features)
(PACA_AU_array)

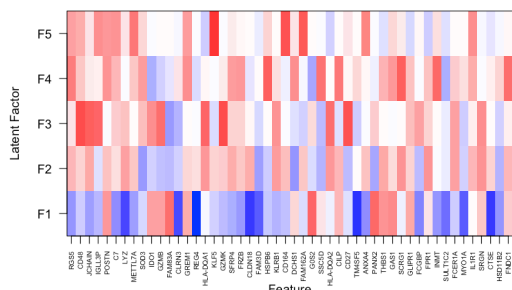


Figure 4: GEP Feature Weights (Top 50 Features)
(PACA_AU_seq)

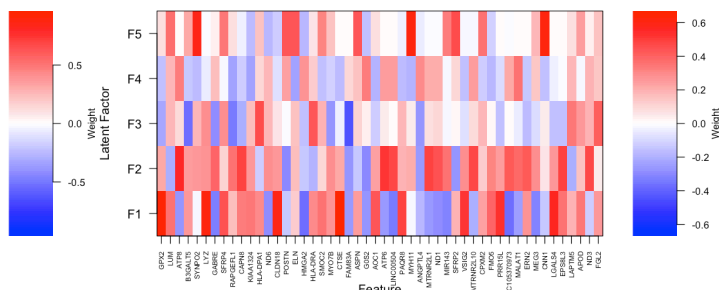


Figure 19: PACA AU array GEP heatmap

Figure 20: PACA AU seq GEP heatmap

Figure 4: GEP Feature Weights (Top 50 Features)
(Puleo_array)

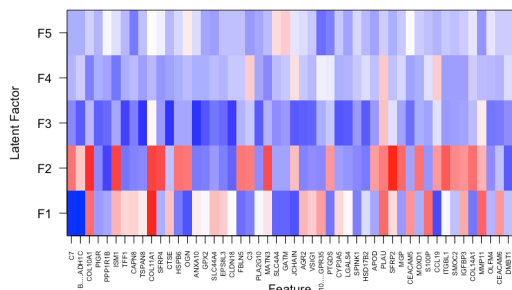


Figure 21: Puleo array GEP heatmap

3.7.5 Kaplan-Meier Curves

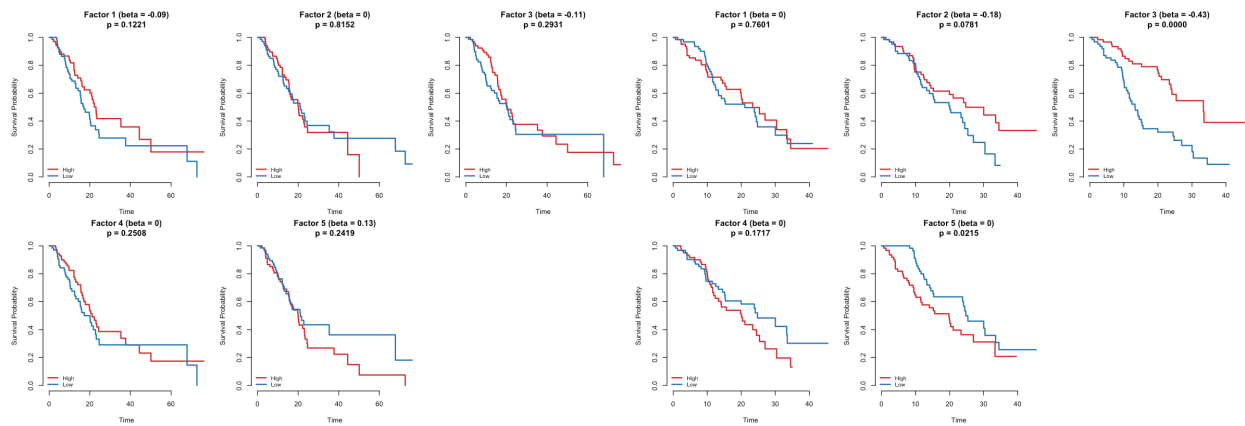


Figure 22: TCGA PAAD Kaplan-Meier

Figure 23: CPTAC Kaplan-Meier

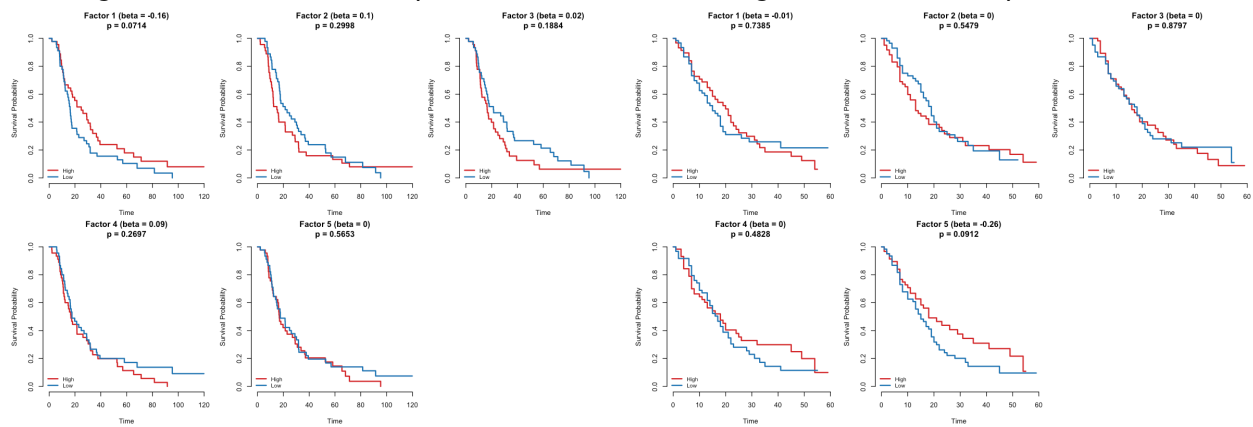


Figure 24: Dijk Kaplan-Meier

Figure 25: Moffitt GEO array Kaplan-Meier

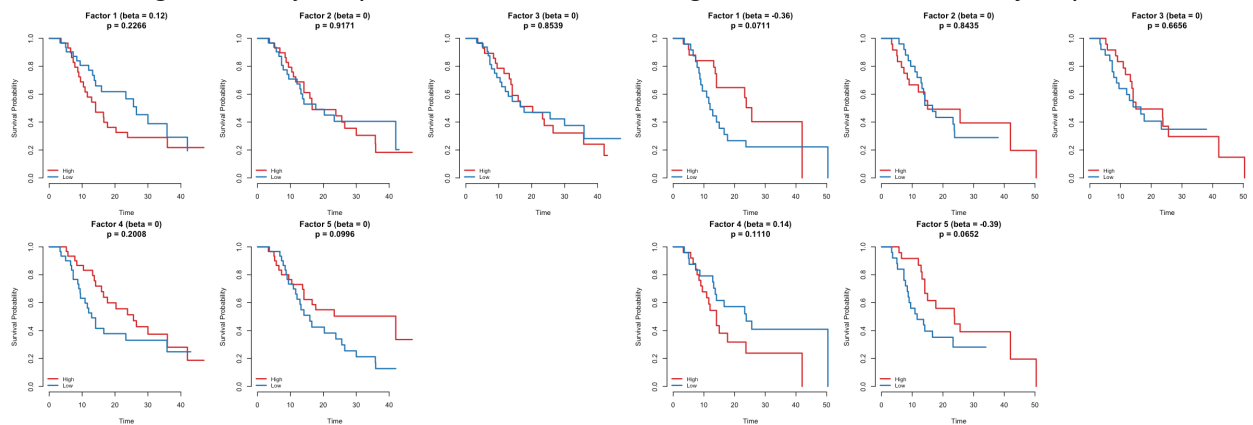
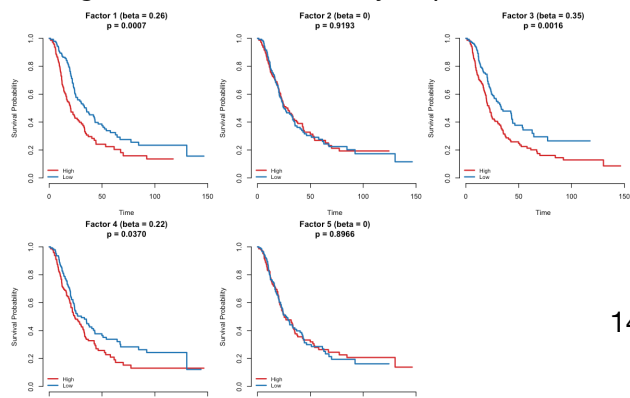


Figure 26: PACA AU array Kaplan-Meier

Figure 27: PACA AU seq Kaplan-Meier



4 Cross-Dataset Performance Summary

Table 10: Performance by platform. Δ C-index = Supervised – PCA.

Dataset	Platform	n	Iters	RMSE	C (PCA)	C (Supervised)	Δ C
Microarray							
Moffitt_GEO_array	Microarray	123	14	0.725	0.571	0.593	0.022
PACA_AU_array	Microarray	63	83	1.016	0.653	0.623	-0.030
Puleo_array	Microarray	288	72	0.588	0.664	0.650	-0.014
Proteomics							
CPTAC	Proteomics	129	259	1.263	0.703	0.691	-0.012
RNA-seq							
Dijk	RNA-seq	90	120	1.122	0.610	0.622	0.012
PACA_AU_seq	RNA-seq	52	186	0.707	0.776	0.788	0.012
TCGA_PAAD	RNA-seq	144	171	1.204	0.640	0.620	-0.020

Platform-level observations:

- **Microarray** (n = 63–288): Fast convergence (14–83 iters), low RMSE (0.59–1.02). Puleo_array is the standout — large n, strong signal, 3 prognostic factors.
- **RNA-seq** (n = 52–144): Slower convergence (120–186 iters), RMSE ~1.1–1.2. PACA_AU_seq is the highest-C-index cohort despite the smallest RNA-seq n.
- **Proteomics** (CPTAC, n = 129): Slowest to converge (259 iters), highest RMSE (1.26). Two strong prognostic factors ($\beta_3 = -0.435$, log-rank p < 0.001).

5 Cross-Platform Comparison: PACA_AU Pair

The PACA_AU cohort is profiled by both microarray (PACA_AU_array, n=63) and RNA-seq (PACA_AU_seq, n=52), providing a rare within-cohort cross-platform view.

Observations:

- Both platforms identify Factor 1 as the dominant variance-explaining factor (PVE ~8–13%).
- PACA_AU_array yields one weakly prognostic factor ($\beta_1 = 0.12$); PACA_AU_seq yields three factors with non-zero β , two of which are negative, suggesting the RNA-seq data captures a richer prognostic signal despite smaller n.
- C-index: array = 0.653 (PCA) vs 0.623 (supervised); seq = 0.776 (PCA) vs 0.788 (supervised) — supervised wins only on the seq side.
- Without sample-level alignment across the two assays, direct factor correspondence cannot be established; this would require paired samples.

Table 11: PACA_AU array vs. seq: factor summary comparison.

Dataset	Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
PACA AU array (Microarray)					
PACA_AU_array	1	0.120	0.2266	100	7.77
PACA_AU_array	2	0.000	0.9171	100	6.88
PACA_AU_array	3	0.000	0.8539	100	2.46
PACA_AU_array	4	0.000	0.2008	100	2.68
PACA_AU_array	5	0.000	0.0996	100	2.81
PACA AU seq (RNA-seq)					
PACA_AU_seq	1	-0.357	0.0711	100	12.55
PACA_AU_seq	2	0.000	0.8435	100	7.76
PACA_AU_seq	3	0.000	0.6656	100	3.52
PACA_AU_seq	4	0.143	0.1110	100	2.90
PACA_AU_seq	5	-0.395	0.0652	100	2.89

6 Within-Modality Comparisons

6.1 RNA-seq Trio (TCGA_PAAD, Dijk, PACA_AU_seq)

Table 12: RNA-seq cohort comparison.

Dataset	n	Censoring %	Iters	RMSE	C (PCA)	C (Supervised)	ΔC
TCGA_PAAD	144	47.9	171	1.204	0.640	0.620	-0.020
Dijk	90	10.0	120	1.122	0.610	0.622	0.012
PACA_AU_seq	52	40.4	186	0.707	0.776	0.788	0.012

- PACA_AU_seq has the highest C-index and the highest event rate (60% events, 10% censoring in Dijk is the extreme — nearly all events observed).
- TCGA_PAAD converges slower (171 iters) partly because n=144 with 48% censoring provides a weaker Cox likelihood signal per iteration.

6.2 Microarray Trio (Moffitt_GEO_array, PACA_AU_array, Puleo_array)

- Moffitt_GEO_array converges in just 14 iterations — the smoothest landscape of any cohort.
- Puleo_array (n=288, the largest cohort) is the only dataset with multiple strongly significant prognostic factors ($\beta_1, \beta_3, \beta_4$ all non-zero with log-rank p < 0.05), reflecting the greater power available with larger n.

Table 13: Microarray cohort comparison.

Dataset	n	Censoring %	Iters	RMSE	C (PCA)	C (Supervised)	ΔC
Moffitt_GEO_array	123	32.5	14	0.725	0.571	0.593	0.022
PACA_AU_array	63	39.7	83	1.016	0.653	0.623	-0.030
Puleo_array	288	37.2	72	0.588	0.664	0.650	-0.014

7 CPTAC (Proteomics)

CPTAC is the only non-transcriptomic cohort: protein abundance measured by mass spectrometry. It required the most iterations (259) to converge, likely reflecting the fundamentally different variance structure of protein abundance compared to transcript counts.

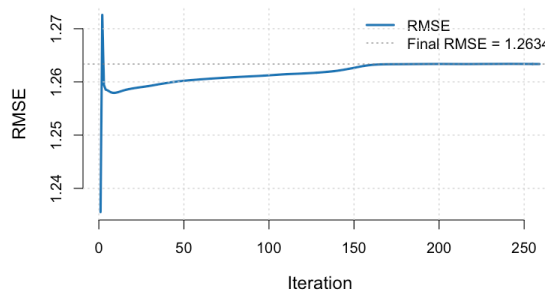
Table 14: CPTAC factor summary.

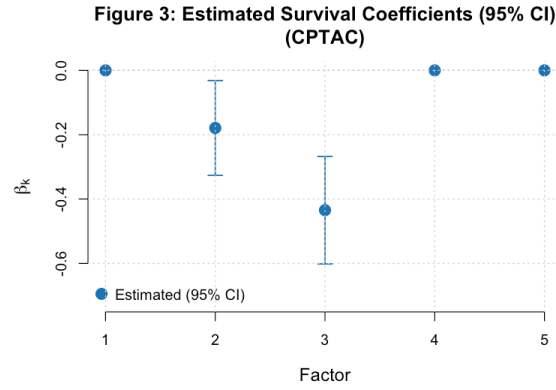
Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	0.000	0.7601	100	12.31
2	-0.179	0.0781	100	4.38
3	-0.435	0.0000	100	3.80
4	0.000	0.1717	100	2.05
5	0.000	0.0215	100	5.27

Observations:

- Factor 3 is the dominant prognostic program: $\beta_3 = -0.435$, log-rank $p < 0.001$. Patients with high loading on this factor have substantially worse survival.
- Factor 2 is borderline significant ($p = 0.078$) with $\beta_2 = -0.179$.
- Factor 5 shows significant log-rank separation ($p = 0.022$) but $\beta_5 = 0$ — suggesting a non-linear or non-proportional survival relationship that the Cox model shrinks to zero.
- C-index: PCA = 0.703, Supervised = 0.691; supervised is slightly lower, possibly because the point-normal prior is conservative on small prognostic signals.

Figure 1: Reconstruction RMSE Across CAVI Iterations (CPTAC)





8 Horizontal Integration: Pooled RNA-seq

Three RNA-seq cohorts (TCGA_PAAD, Dijk, PACA_AU_seq) share 1,573 common genes after top-5,000 variance filtering. These were pooled into a single matrix ($n = 286$) and fit with a single SBMF model.

Table 15: Pooled RNA-seq fit: factor summary ($n=286$, $p=1573$ common genes, $K=5$).

Factor	$\hat{\beta}$	Log-rank p	Non-zero %	PVE %
1	0.000	0.9078	100	16.10
2	0.113	0.0263	100	9.83
3	-0.205	0.0217	100	6.44
4	-0.182	0.0226	100	4.68
5	0.000	0.6196	100	3.53

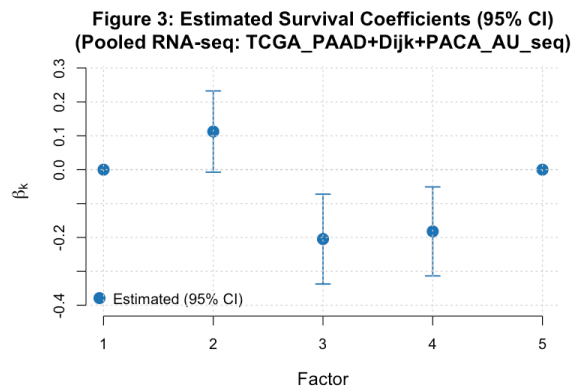
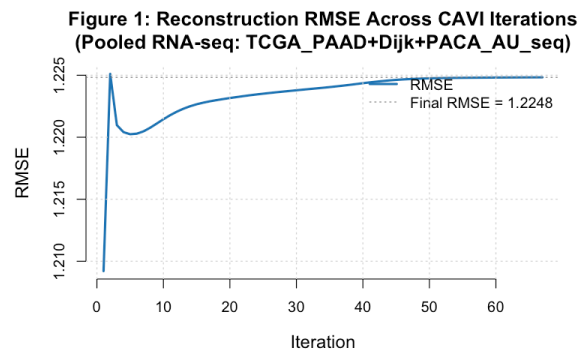
Table 16: Sample composition of the pooled RNA-seq dataset.

Cohort	N samples
Dijk	90
PACA_AU_seq	52
TCGA_PAAD	144

Observations:

- The pooled model converges quickly (67 iterations, RMSE = 1.225), suggesting the shared gene set provides a coherent signal.
- Factors 2, 3, 4 are prognostic ($\beta \neq 0$, log-rank $p < 0.03$); Factors 1 and 5 are null.
- Factor 1 (PVE = 16.1%) likely captures cohort-level mean differences — the largest single source of variance in a pooled, non-batch-corrected matrix.

- **Caveat:** No batch correction was applied. Factor 1 and possibly other factors may partially reflect study-specific technical variation rather than biology. Future work: apply ComBat or similar before pooling, or include cohort as a covariate in the Cox model.



9 Conclusion & Future Directions

The SBMF model successfully fits all 7 PDAC cohorts across three assay platforms, recovering sparse prognostic gene expression programs in each. Key results:

Cohort	Platform	Strongest finding
Puleo_array	Microarray	3 prognostic GEPs (n=288); largest single-dataset signal
CPTAC	Proteomics	Factor 3: $\beta = -0.435$ ($p < 0.001$); proteomic programs drive survival
PACA_AU_seq	RNA-seq	Highest overall C-index (0.788); 3 non-zero β factors
Dijk	RNA-seq	Nearly fully observed (10% censoring); Factor 1 dominant

Immediate next steps:

1. **K selection via cross-validation** — currently fixed at $K = 5$ for all cohorts; optimal K likely varies by platform and sample size. Longleaf HPC recommended for the CV grid search.
2. **Batch correction before pooling** — apply ComBat to the RNA-seq trio before pooling to separate biological from technical factors.
3. **Top-gene interpretation** — run gene set enrichment analysis (GSEA/fgsea) on the top-loaded genes per factor to assign biological pathway labels to each GEP.
4. **Full ELBO tracking** — the current ELBO proxy tracks only the genomics log-likelihood; adding the Cox likelihood and KL terms would give a proper variational lower bound for model comparison across K .
5. **Vertical multi-modal integration** — for cohorts with paired RNA + protein data (e.g., CPTAC has both transcriptomics and proteomics), a shared- L model would allow factors to explain variance in both modalities simultaneously.